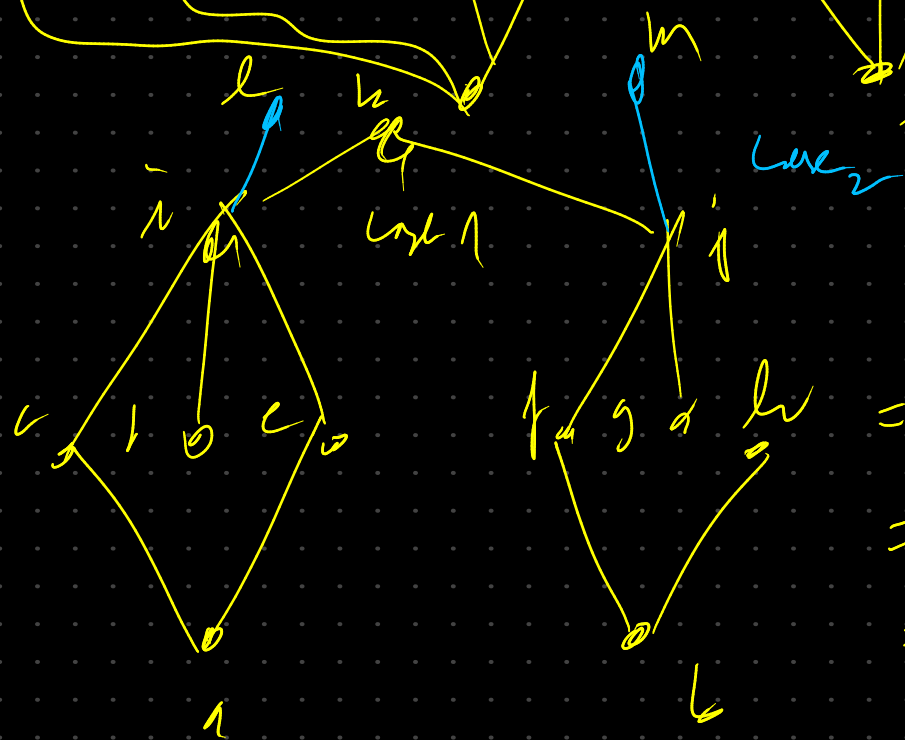
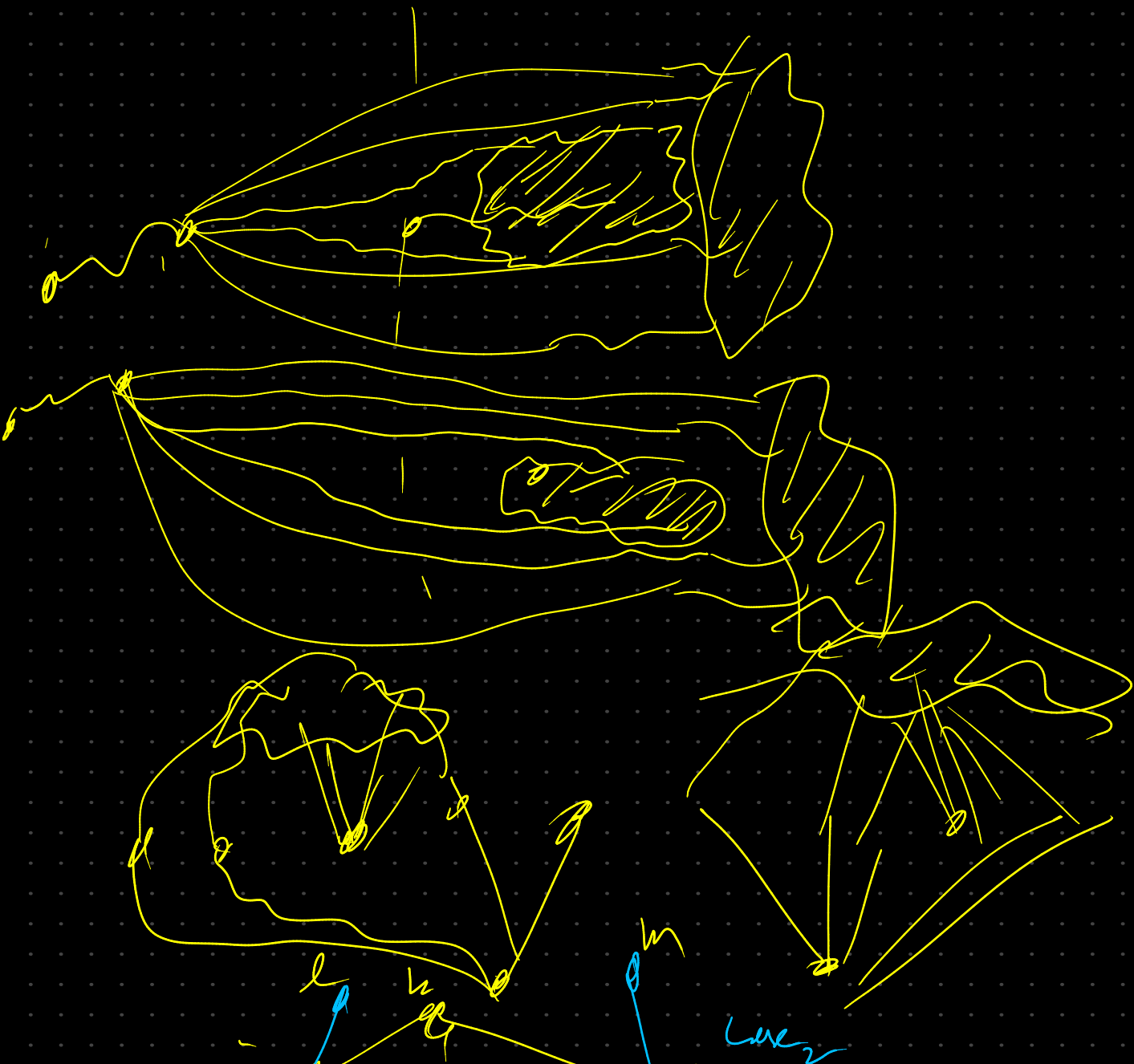
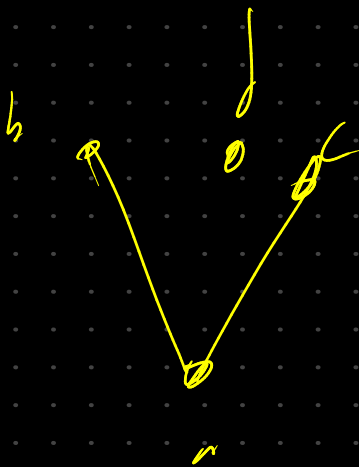


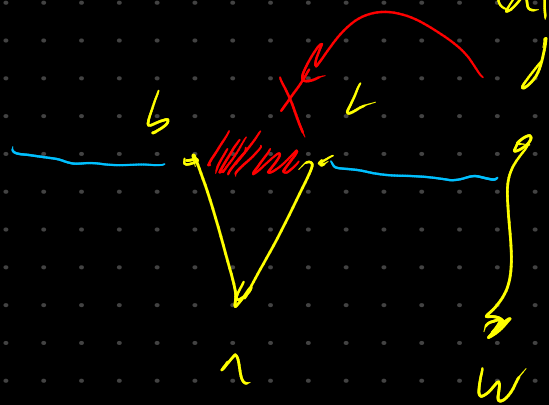


$$\begin{aligned}
 \varphi(a,b) &= \varphi(i,j) \\
 &= \varphi(c,f) = \varphi(d,g,h) \\
 &= \varphi(c,g) \\
 &= \varphi(c,h) \\
 &= \varphi(e,f,g,h)
 \end{aligned}$$

the issue comes from the non-independent edges (edges that share an endpoint), because it's components are not taking in consideration in the 2-SAT, the case that leads to issues is defining the order of the non-shared endpoints to each other and to vertices that don't share an edge that is independent to the 2 edges that are internal but not ind.

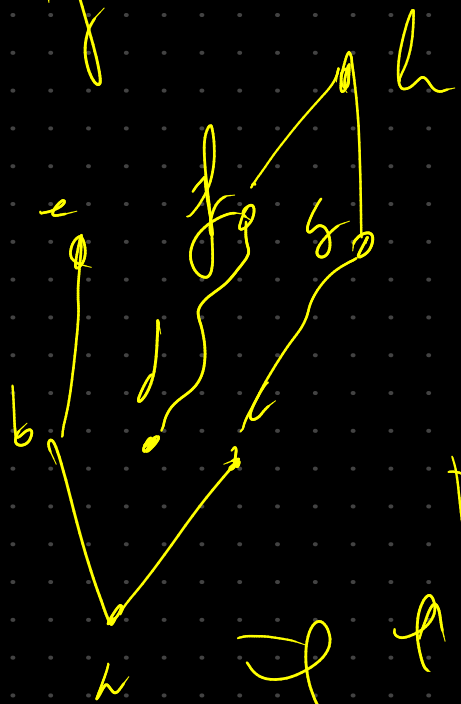


if I have $e = (b, d) \in E(b)$
and e and (a, c) are ind.



$$\begin{aligned} \varphi(b, d) \\ &= \varphi(c, d) \\ &= \varphi(a, c) \end{aligned}$$

st if it don't have e u



we can (not repeat) the transitivity
change without denying the
transitivity constraint.

$$P \quad P \left(\begin{array}{ccc} b & d \\ \frac{11}{1} & - & \frac{11}{1} \\ & \frac{11}{1} & \end{array} \right) = P(d, c) = P \left(\begin{array}{ccc} b & c \\ \frac{11}{1} & - & \frac{11}{1} \\ & \frac{11}{1} & \end{array} \right)$$

$$P(b, c) = 1 = P(e, g) = 1$$

$$P \left(\begin{array}{ccc} b & d \\ \frac{11}{1} & - & \frac{11}{1} \\ & \frac{11}{1} & \end{array} \right) = P \left(\begin{array}{ccc} f & g \\ \frac{11}{1} & - & \frac{11}{1} \\ & \frac{11}{1} & \end{array} \right)$$

$$P(e, f) = P(b, d) = 0$$

(if we have less than
3 q. classes in
each connected comp. ✓)

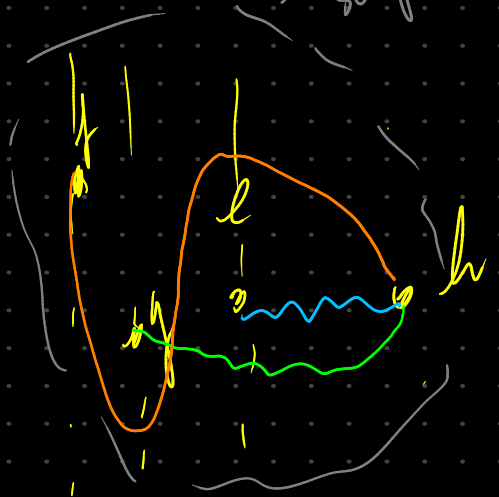
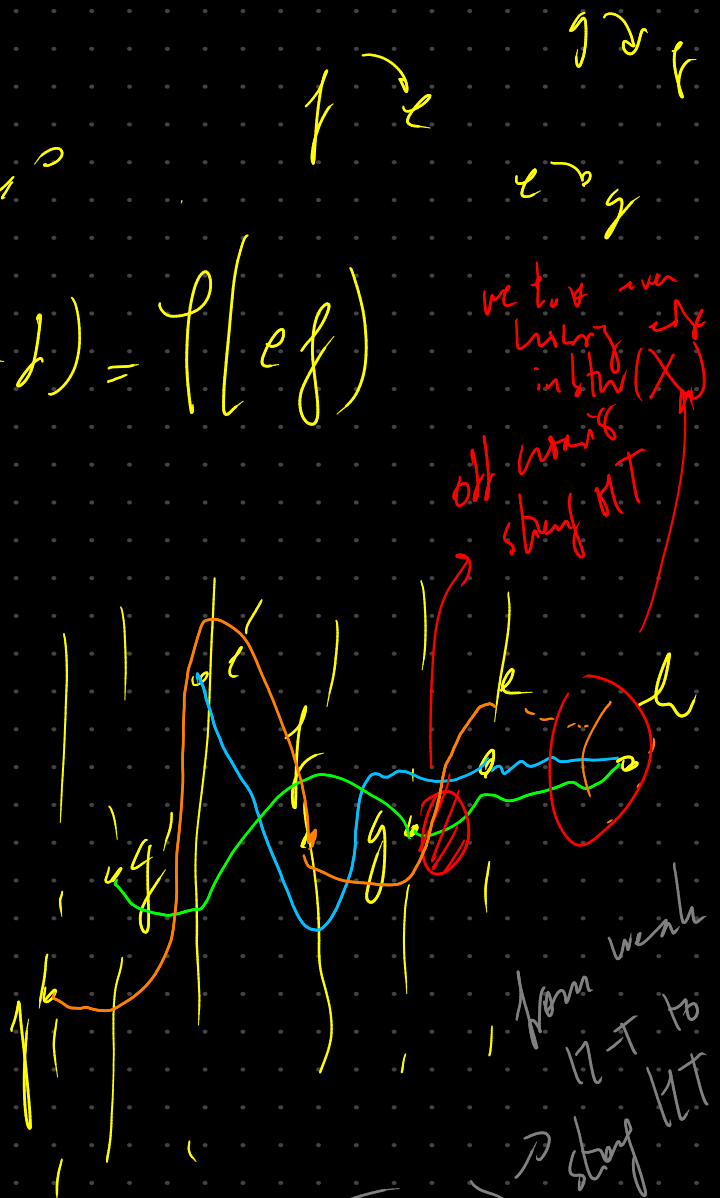


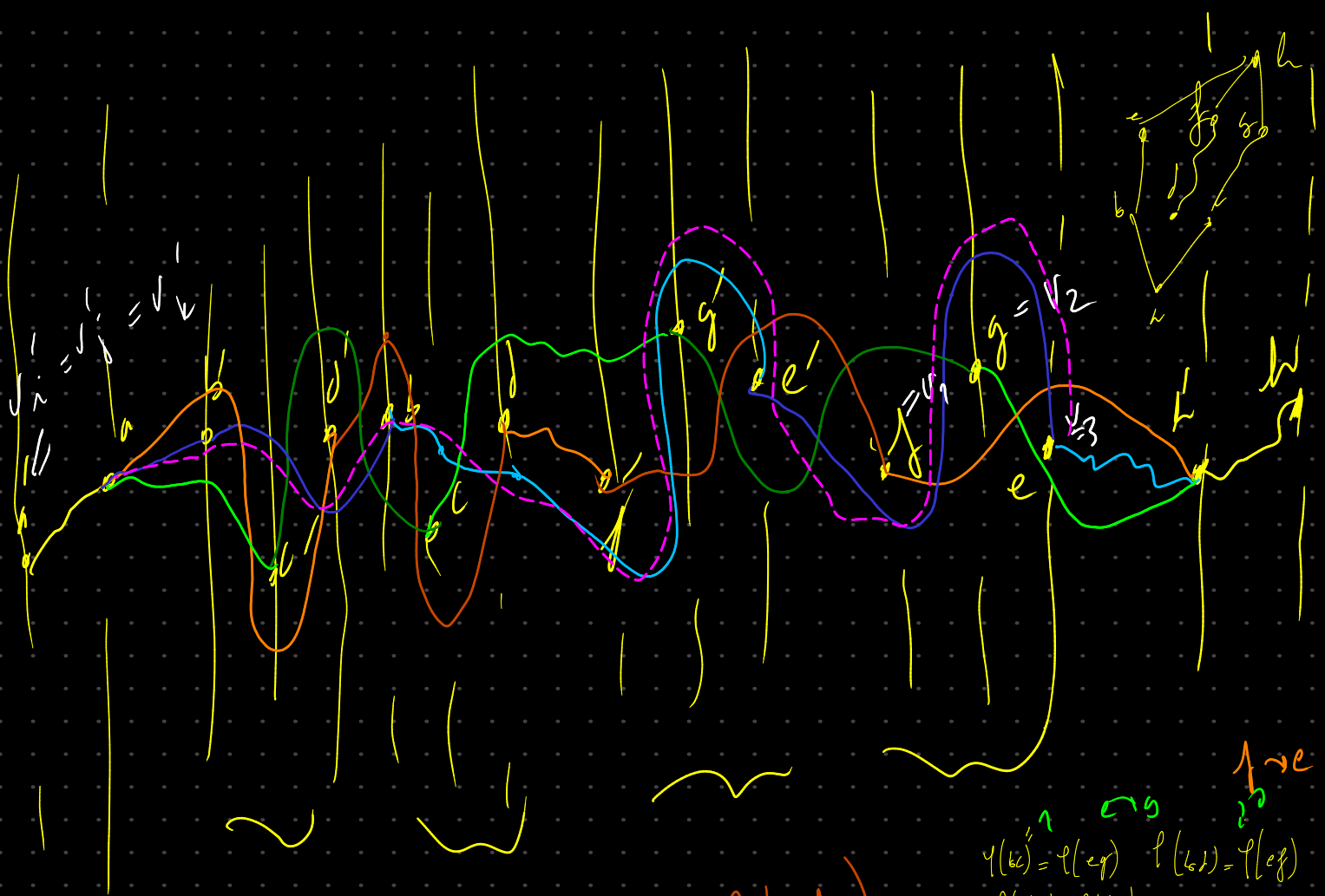
circular relation
between v. of
edges that are adjacent

$$\gamma(bc) = \ell(e_g) \quad \ell(bc) = \ell(e_g)$$

$$\ell(bc) = \ell(fg)$$

this is the second
case where it could
lead to prove that's
not possible to have
planar emb. with ℓ



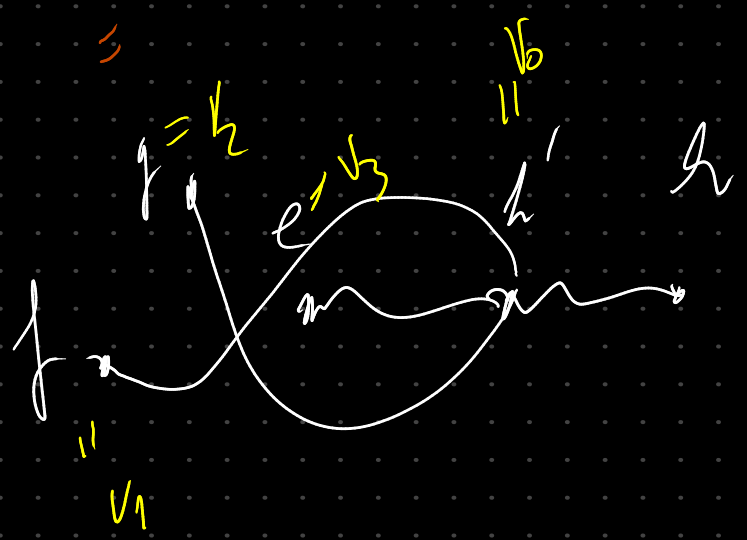


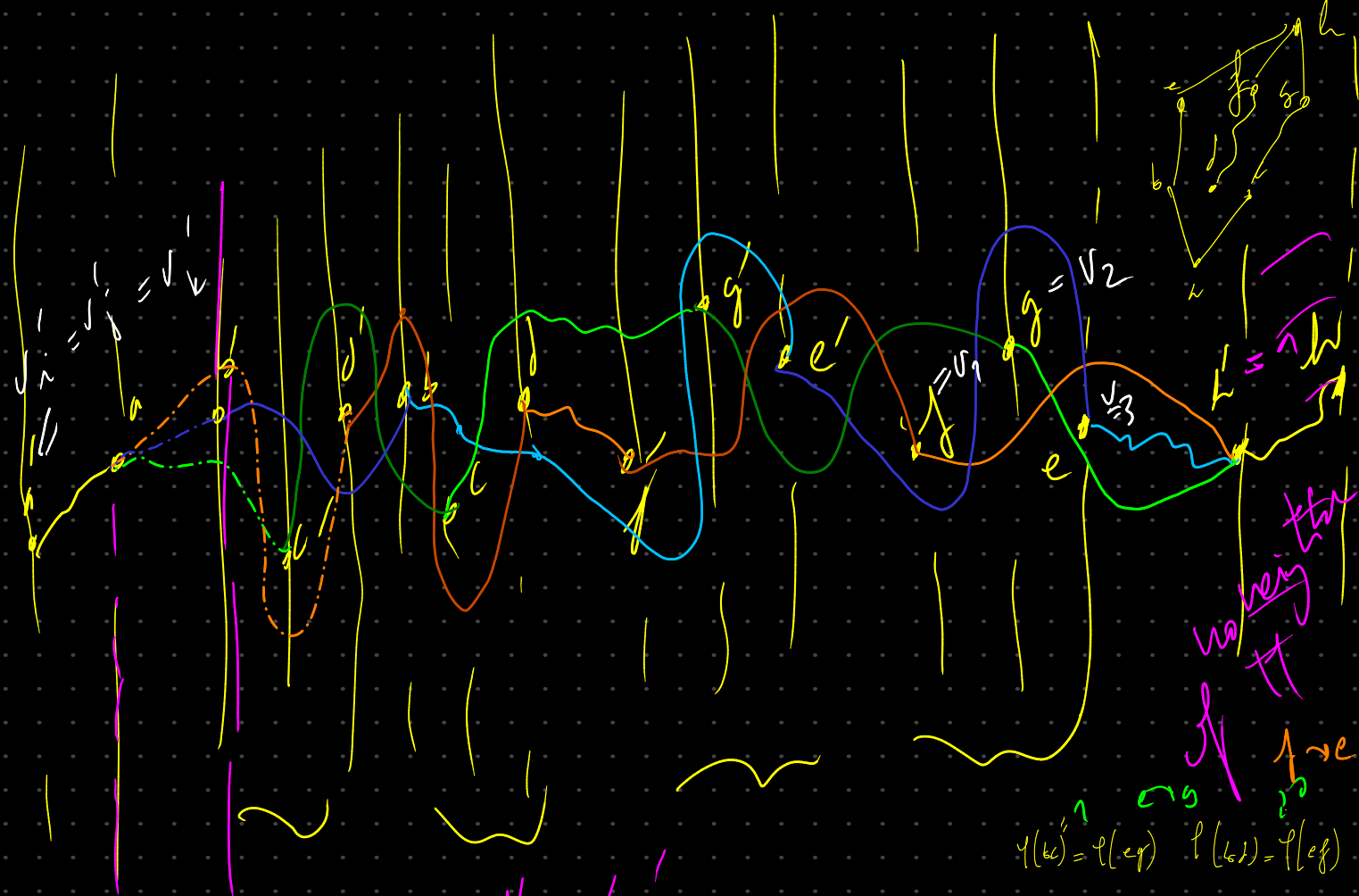
$$p(\delta b) = 0$$

$$p(\delta a)$$

$$p(\delta c) = p(\delta d)$$

$$p(\delta e) = p(\delta f)$$





$$f(h) = f(e) \quad f(h) = f(e)$$

$$H = h_1$$

$$N(H) = \{x, y, z, w, v, u, t, s, r, q, p, o, n, m, l, k, j, i, h, g, f, e, d, c, b, a\}$$

$$V(H) = \{x, y, z, w, v, u, t, s, r, q, p, o, n, m, l, k, j, i, h, g, f, e, d, c, b, a\}$$

$$H = h_1$$

$$N(H) = \{x, y, z, w, v, u, t, s, r, q, p, o, n, m, l, k, j, i, h, g, f, e, d, c, b, a\}$$

$$V(H) = \{x, y, z, w, v, u, t, s, r, q, p, o, n, m, l, k, j, i, h, g, f, e, d, c, b, a\}$$

